

Advanced (Habanero)**Q1.** Which of the following is a function?

- (A) $y = x^2$
 (B) $x = y^2$
 (C) $y^2 = x^2$
 (D) $x^2 + y^2 = 1$
 (E) $y = |x|$

Q2. What is the domain of the function $f(x) = \frac{1}{x}$?

- (A) $(-\infty, \infty)$
 (B) $(-\infty, 0) \cup (0, \infty)$
 (C) $[0, \infty)$
 (D) $(-\infty, 0)$
 (E) $[1, \infty)$

Q3. Determine if the relation $\{(1, 2), (2, 3), (3, 2), (4, 3)\}$ is a function.

- (A) Yes
 (B) No
 (C) Maybe
 (D) Depends on the context
 (E) Insufficient information

Q4. What is the range of the function $f(x) = x^2 + 1$?

- (A) $[1, \infty)$
 (B) $(-\infty, 1]$
 (C) $[0, \infty)$
 (D) $(-\infty, 0)$
 (E) $(-1, 1)$

Q5. Which of the following is an example of function notation?

- (A) $y = x^2$
 (B) $f(x) = x^2$
 (C) $x = y^2$
 (D) $y^2 = x^2$
 (E) $x^2 + y^2 = 1$

Q6. What is the value of $f(-2)$ if $f(x) = x^2 - 3$?

- (A) -7
 (B) -1
 (C) 1
 (D) 5
 (E) 7

Q7. Determine the domain of the function $f(x) = \sqrt{x-1}$.

- (A) $(-\infty, 1)$
 (B) $[1, \infty)$
 (C) $(-\infty, 0)$
 (D) $[0, 1]$
 (E) $(-1, 1)$

Q8. What is the range of the function $f(x) = \sqrt{x}$?

- (A) $[0, \infty)$
 (B) $(-\infty, 0)$
 (C) $(-\infty, \infty)$
 (D) $(-1, 1)$
 (E) $[1, \infty)$

Open Ended Questions (FRQ)**Q9.** Given the function $f(x) = x^2 - 2x - 3$, find the domain and range of the function and evaluate $f(2)$.**Q10.** Determine if the relation $\{(1, 1), (2, 2), (3, 3), (4, 4)\}$ is a function and explain why or why not. If it is a function, find its domain and range.**Chef's Tips**

- Read each question carefully.
- Show all work and explain your reasoning.
- Use proper function notation (e.g., $f(x)$).